MAT2002 **Name (Print):** _____
Spring 2026
HW3
Deadline: 15th March

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1. Solve the IVP: $y'' + 2y' + 2y = 0, \quad y(\pi/4) = 2, \quad y'(\pi/4) = -2$

2. Solve the IVP: $y'' + y = 0, \quad y(\pi/3) = 2, \quad y'(\pi/3) = -4$

3. Solve the following for $t > 0$:

   (i) $t^2y'' + 3ty' - 3y = 0$

   (ii) $t^2y'' + 7ty' + 10y = 0$

4. Solve the following IVP:

   (i) $9y'' + 6y' + 82y = 0, \quad y(0) = -1, \quad y'(0) = 2$

   (ii) $y'' + 4y' + 4y = 0, \quad y(-1) = 2, \quad y'(-1) = 1$

5. In each of following, use the method of reduction of order to find a second solution of the given differential equation:

   (i) $xy'' - y' + 4x^3y = 0, \quad x > 0; \quad y_1(x) = \sin x^2$

   (ii) $(x-1)y'' - xy' + y = 0, \quad x > 1; \quad y_1(x) = e^x$

6. Find the general solution for : $y'' - y' - 2y = \cosh 2t$
   Hint: $\cosh t = \left(e^t + e^{-t}\right)/2$

7. Solve the IVP $y'' - 2y' - 3y = 3te^{2t}, \quad y(0) = 1, \quad y'(0) = 0$

8. Consider the equation
$$y'' - 3y' - 4y = 2e^{-t} \tag{1}$$
   $y_1(t) = e^{-t}$ and $y_2(t) = e^{4t}$ are solutions of the corresponding homogeneous equation. Adapting the method of reduction of order , seek a solution of the nonhomogeneous equation of the form $Y(t) = v(t)y_1(t) = v(t)e^{-t}$, where $v(t)$ is to be determined.

   (i) Substitute $Y(t), Y'(t)$, and $Y''(t)$ into Eq. (1) and show that $v(t)$ must satisfy $v'' - 5v' = 2$.

   (ii) Let $w(t) = v'(t)$ and show that $w(t)$ must satisfy $w' - 5w = 2$. Solve this equation for $w(t)$.

   (iii) Integrate $w(t)$ to find $v(t)$ and then show that
$$Y(t) = -\frac{2}{5}te^{-t} + \frac{1}{5}c_1e^{4t} + c_2e^{-t}$$
   The first term on the right side is the desired particular solution of the nonhomogeneous equation. Note that it is a product of t and e^{-t}.

9. Determine the general solution of

$$y'' + \lambda^2 y = \sum_{m=1}^{N} a_m \sin m\pi t$$

where $\lambda > 0$ and $\lambda \neq m\pi$ for $m = 1, \ldots, N$.

10. In each of following, verify that the given functions y_1 and y_2 satisfy the corresponding homogeneous equation; then find a particular solution of the given nonhomogeneous equation.

(i) $t^2y'' - 2y = 3t^2 - 1, \quad t > 0; \quad y_1(t) = t^2, \quad y_2(t) = t^{-1}$

(ii) $ty'' - (1+t)y' + y = t^2e^{2t}, \quad t > 0; \quad y_1(t) = 1 + t, \quad y_2(t) = e^t$